

Lab - FHIRPath and Search operations

Patrick W. Jamieson, M.D.
Technical Product Manager

Russ Leftwich, M.D.
Senior Clinical Advisor, Interoperability



Objectives



Applying Knowledge of FHIR search



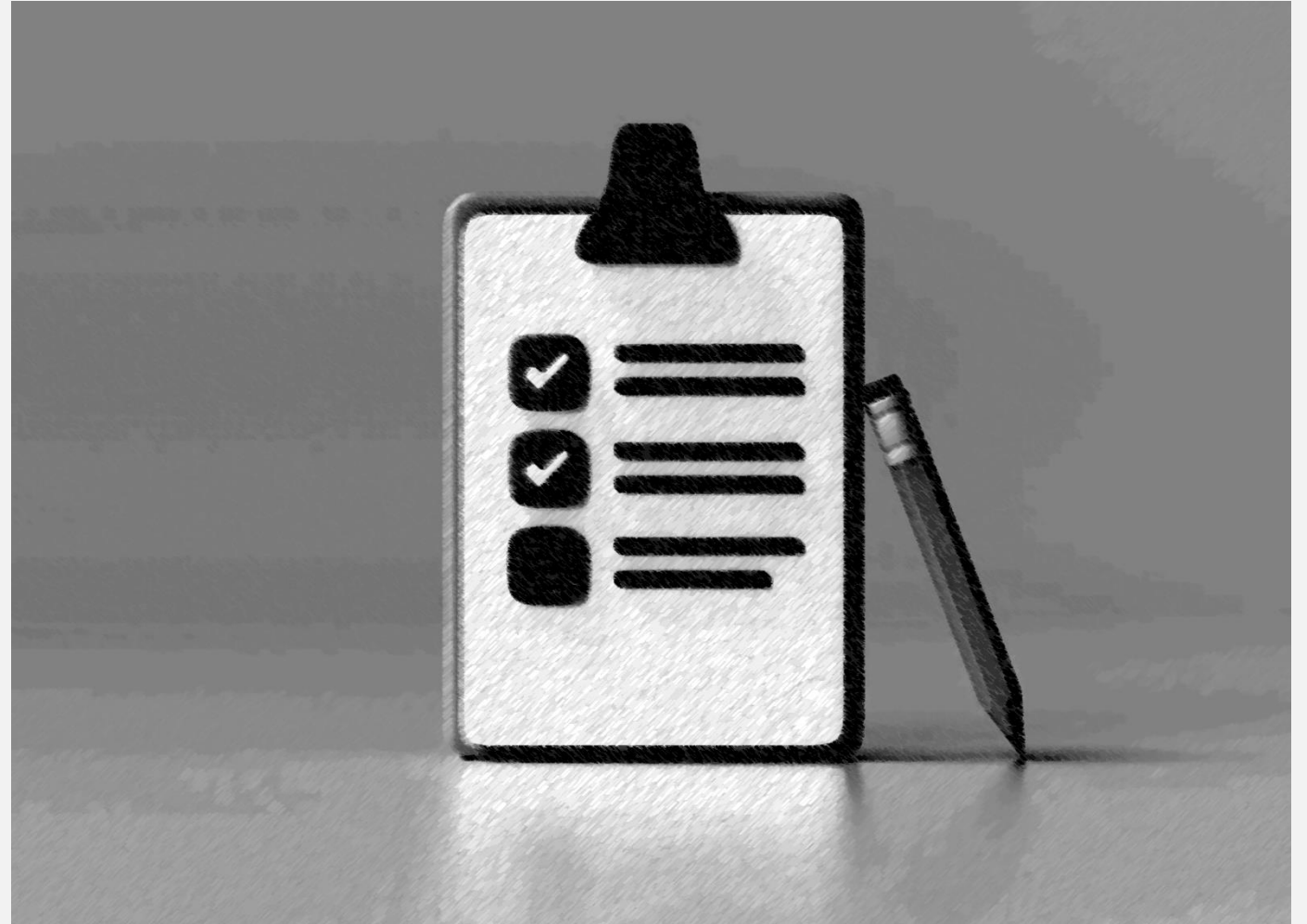
Extracting specific elements from FHIR search bundles using FHIRPath



Translating Clinical Questions into FHIR commands (combination of FHIR Search and FHIRPath)

Quick Review

- FHIRPath
- FHIR Search



Basic FHIR Expression Examples



Extracting a Patient's Given Name

Expression: Patient.name.given

Example Use: Retrieve a patient's first name from their record for a personalized greeting in a patient portal.

Checking for the Existence of a Birth Date

Expression: Patient.birthDate.exists()

Example Use: Validate that the birth date field is populated before scheduling age-dependent clinical procedures.

Filtering by Gender

Expression: Patient.gender = 'female'

Example Use: Retrieve records for female patients to support a study on gender-specific treatment outcomes.

Retrieving Medication Dosage

Expression:
MedicationRequest.dosageInstruction.doseQuantity

Example Use: Extract dosage instructions from a medication request to verify correct administration in clinical settings.

Common Search Parameters



Search Parameter	Description
_id	Search by resource ID (e.g., _id=12345)
name	Search by patient or practitioner name (e.g., name=Smith)
identifier	Search by unique identifier (e.g., GET /Patient?identifier=http://hospital.org/mrn 12345)
code	Search by coded values (e.g., code=http://loinc.org 85354-9)
date	Search by date (e.g., date=ge2024-01-01)

Common Search Parameters (Continued)



Search Parameter	Description
<code>_include</code>	Include referenced resources (e.g., <code>_include=Encounter.patient</code>)
<code>_revinclude</code>	Find resources that reference a resource (e.g., <code>_revinclude=Observation:subject</code>)
<code>_count</code>	Limit the number of results returned (e.g., <code>_count=10</code>)
<code>_sort</code>	Sort results by a field (e.g., <code>_sort=name</code>)
<code>_has</code>	Reverse chaining search (e.g., <code>_has:Observation:subject:code=1234</code>)

FHIR Headers We Will Use



Header	Purpose	Why It's Important
Authorization: Basic _System:ISCDEMO	Authentication	Prevents unauthorized access
Accept: */*	Response format	Ensures proper content handling
Content-Type: application/fhir+json application/json-patch+json	Request body format, slightly different for PATCH	Specifies the payload as FHIR JSON
Accept-Encoding: gzip, deflate, br	Compression preferences	Optimizes response size & speed
Prefer: return=representation	Response detail level	Returns the full created resource

Instructions



Try each challenge
yourself first

Watch the video
solution for confirmation
or if you get stuck



Background



Nurse Emily Carter, a dedicated diabetes specialist at Fairway Hospital, has been tasked with a critical mission—identifying all patients at risk of developing Type 2 Diabetes. With an increasing number of patients showing borderline HbA1c levels (5.7% - 6.4%), the hospital's preventive care team wants to intervene early. Emily knows that early detection and lifestyle modifications can prevent many from progressing to full-blown diabetes, but with hundreds of patients coming through the hospital, manually reviewing charts isn't feasible.

Determined to use **FHIR-powered data retrieval**, Emily decides to run a search query on the hospital's **FHIR server**. By pulling all **patients with recent HbA1c test results in the prediabetes range**, she can quickly generate a **targeted list of at-risk individuals**. With this data, she plans to schedule follow-up visits, connect patients with dietitians, and enroll them in a **preventive wellness program**. This is more than just a search—it's a chance to change lives before diabetes takes hold.

Problem One



- Emily first tries to find all the patients in the system's FHIR repository with the condition of prediabetes. She happens to know that prediabetes is a concept in SNOMED CT with a value of 15777000
- Write a FHIR Search Query that will retrieve all the patients with prediabetes.
- How many patients did you find?

Problem Two



Emily wants to explore how many Hemoglobin A1c Observations occur in a typical patient with prediabetes, so she chooses the first patient in her searchset for problem one, Mrs. Gulgowski (Patient/12251).

- Write a FHIR search query to determine how many Hemoglobin A1c Observations have been done on this Patient (Patient/12251)?
- How many Observations did you find?

Problem Three



Emily would like to get a printout of all the Hemoglobin A1c values on Mrs. Gulgowski. Emily knows that once she gets the searchset bundle from problem two, it will be an easy job to extract all the values using FHIRPath.

- Use the Jupyter notebook to execute the first few cells to set up making a FHIR search request from Python. In cell 6, you will need to plug your FHIR search query from problem two into the URL to retrieve all the observations using Python
- In cell nine create a FHIRPath expression to get all the HgbA1c values. Using Python create an array of all the values.
- In cell 10 create a sorted array of values. For this problem you will need to use Python because the FHIRPath sort() function does not work with this FHIRPath client library.

Problem Four



Emily has an idea for finding all patients with borderline HgbA1c. She wants to get all the patients with borderline or high HgbA1c values, but first she wants to check her thinking on just one patient, Mrs. Gulgowski.

- Write a FHIR search rest query to obtain all the HgbA1c values of Mrs. Gulgowski (Patient/12251), which are greater than or equal to 6.0 mg/dl
- How many values did you get?

Problem Five



Emily now is going to find all patients with elevated HgbA1c values..

- Write a FHIR search rest query to obtain all the patients with HgbA1c values which are greater than or equal to 6.0 mg/dl
- How many values did you get?
- Did you find the same number, or more or less?

Problem Six



Emily would like to compare the two list of patients she retrieved and identify who was discovered using HgbA1c ≥ 6.0 mg/dl versus the condition of prediabetes.

- Using the Jupyter notebook take the FHIR search URL you constructed to find all the patients with prediabetes and write a FHIRPath expression (and Python) to create a set of all FHIR ids.
- Do the same for patients with HgBA1c ≥ 6.0 mg/dl
- Use the enclosed Python function to compare the two sets? What did you find?

PreventHF: A Home Health Startup for Heart Failure Prevention



As hospital readmission rates for congestive heart failure (CHF) patients continue to rise, a new startup, PreventHF, is on a mission to change the landscape of home healthcare. Founded by a passionate team of healthcare professionals and engineers, PreventHF specializes in remote patient monitoring, aiming to catch early warning signs of heart failure before they lead to hospitalization.

At the core of PreventHF's innovation is a smart wireless scale, designed for CHF patients to use every morning. This user-friendly scale automatically transmits each patient's weight as a FHIR Observation resource, securely sending data to the company's cloud-based health monitoring platform.

PreventHF - continued



The **PreventHF Nursing Team** actively monitors patient weight trends using **FHIR search and FHIRPath** to detect rapid weight gain—a key indicator of worsening heart failure due to fluid retention. When concerning changes occur, an alert is triggered, allowing nurses to intervene early by adjusting medications or coordinating a telehealth visit with a cardiologist.

As the lead developer at PreventHF, your task is to optimize the FHIR data queries and searches used by nurses to monitor patient weight trends effectively.

Problem One



- Susan has been assigned to watching the weight fluctuations of a particular congestive heart failure patient, John Herrington. Since Susan is new to this job, she first tries to learn about all the FHIR vital sign observations on John Herrington.
- Write a FHIR Search Query that will retrieve all the vital sign observations on John Herrington.
- How many observations did you find?

Problem Two



- Susan realizes that her patient John Herrington (Patient/70255), will have many more observations in the future, and she is just focused on his daily weights.
- Refine your FHIR Search Query, which uses a token search, which will retrieve only the vital sign observations on John Herrington that deal with weight.
- What system and code did you use?

Problem Three



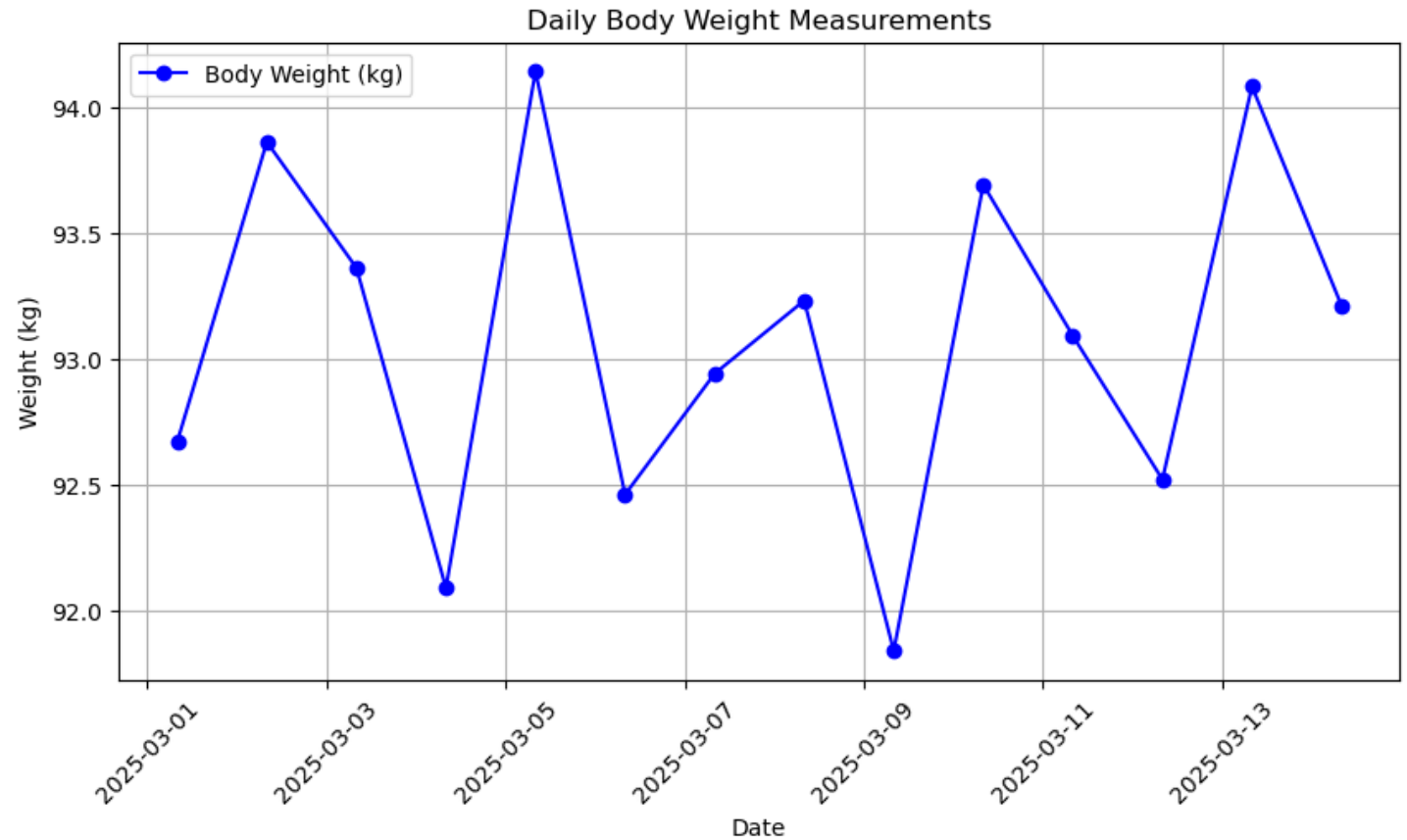
- Susan would like to visualize Mr. Herrington's weight fluctuations. She would like to extract from the FHIR bundle, the weight and date of each observation.
- Construct a FHIRPath expression to extract this information from the search query you built in Problem Two. (Hint: this may take some trial and error but the select operator using a | delimiter between ValueQuantity.value and effectiveDateTime is a workable approach.)



Problem Four

Optional

Construct a plot using matplotlib, it should look something like this.



Problem Five



- Build a simple function to compare the weights on two consecutive days. If the weight increases by more than 1.5 kg create an alert for Susan.
- Apply your function to all the weights in the series.

Additional Patients Needed for Problem Seven



- The last problem requires analyzing patients with surgical conditions. To add these patients, Execute the POST command in fhirsurgbatch.http, a transaction batch file, with the command POST <http://127.0.0.1:8080/csp/healthshare/demo/fhir/r4>. This will populate not only surgical procedures but their associated patients.
- Next execute the POST command in fhirsurgbatchobs.http, a simple batch file, with the command POST <http://127.0.0.1:8080/csp/healthshare/demo/fhir/r4>. This file contains additional observations on these surgical patients.
- These commands only need to be executed once.

Thank you.

